

The effect of transcranial direct current stimulation during motor sequence learning on retention in older adults: a randomized controlled trial

Britt Vandendoorent¹, Sanne Broeder¹, Moran Gilat¹, Evelien Nackaerts¹, Jean-Jacques Orban de Xivry², Alice Nieuwboer¹

¹ Neurorehabilitation Research Group (eNRGy), Department of Rehabilitation Sciences, KU Leuven, Leuven, Belgium

² Movement Control and Neuroplasticity Research Group, Department of Movement Sciences, KU Leuven, Leuven, Belgium

Correspondence: britt.vandendoorent@kuleuven.be

Abstract (350 words)

Transcranial direct current stimulation (tDCS) of the primary motor cortex could enhance the motor learning process in older adults, but so far existing studies are inconclusive and underpowered. It also remains unclear whether the tDCS effects depend on its montage, as the conventional tDCS montage seemingly produces a relatively non-focal stimulation compared to high-definition tDCS. Therefore, this project examined the added effect of both tDCS montages on the primary motor cortex during motor sequence learning. Fifty-two cognitively-intact older adults (mean age: 68.4 ± 8.0 years; mean Mini Mental State Examination score: 28.8 ± 1.3) practiced a 12-item serial reaction time task while receiving two sessions of 20 minutes of tDCS in a double-blind randomized sham-controlled parallel study with a crossover for montage. Mean washout duration between montages was 25.43 ± 12.59 days. Stimulation intensity was set at 1 mA. We assessed reaction times before, immediately after, 24 hours after and during the two practice sessions supplemented by tDCS or sham. As the primary outcome, we studied mean reaction times on the serial reaction time task 24 hours after the first tDCS-

enhanced practice session as compared to sham. The linear mixed model revealed successful retention (pre-retention: $p < 0.001$ and post-retention: $p=1.000$) of the acquired SRTT performance (pre-post: $p < 0.001$) irrespective of receiving tDCS or sham (group*time: $p = 0.737$). Also, these effects were unaffected by montage (time*montage: $p = 0.691$; group*time*montage: $p = 0.574$). Analogously, the acquisition process was unaffected by tDCS or montage (group*time: $p = 0.672$; time*montage: $p = 0.651$; group*time*montage: $p = 0.653$). Significant time effects showed that participants did react faster during all blocks of the second practice session compared to the last block of the first session (all $p < 0.05$). We conclude that there is no surplus value of adding tDCS during motor sequence learning in older adults. We did find that older people benefited from a second practice session to improve their motor learning. Future research should focus on the effects of multiple sessions of tDCS during motor sequence learning on long-term retention and other consolidation outcomes. As learning impairments are more pronounced in people with Parkinson's disease compared to healthy, benefit from stimulation should be studied in this population.

Keywords: Aging, elderly, tES, transcranial electrical stimulation, non-invasive brain stimulation

1 Introduction

2 The consolidation of sequential motor skills is impaired in older adults [1]. Optimal learning of
3 sequential motor skills is characterized by improvements in spatial and/or temporal accuracy of motor
4 sequences with practice [2]. After this acquisition phase, the sequential motor skill goes through the
5 process of consolidation during a period without practice [3]. When successful, retention is reached as
6 indicated by the persistence of the learning gains over time [3]. As such, retention constitutes a primary
7 goal of rehabilitation helping people to maintain the skills needed to function independently [4].
8 Consolidation deficits have been explained by impaired cortico-striatal function, which is reduced in
9 older adults [1].

10 Motor learning behavior imprints in the involved brain areas as a result of practice [5] and includes
11 changes of neural excitability and plasticity of the primary motor cortex (M1), amongst other regions
12 [6]. Neuroplasticity is the capability of the central nervous system to adapt its structure and function
13 to internal and external stimuli [5]. With advancing age, there is a decline in the neuroplasticity
14 potential [7], pointing to the need to investigate methods that optimize motor learning in older adults.

15 Anodal transcranial direct current stimulation (tDCS) has been proposed as one of these methods to
16 boost motor learning [8]. tDCS comprises an electrical current of 1-4 mA applied to the scalp using at
17 least two electrodes [9, 10]. The conventional tDCS montage (CONV-tDCS) consists of two large
18 rectangular electrodes placed over different cortical areas [11], producing a relatively non-focal
19 stimulation across a more extended area than targeted [12]. Additionally, peak current density is
20 thought to take place in the cortical area between electrodes instead of under the target electrode
21 [12]. In an attempt to improve focality, High-Definition tDCS (HD-tDCS) involving small electrodes in a
22 ring montage, with four electrodes surrounding the central target electrode, was developed [13]. This
23 set-up is presumed to constrain the current flow to the area between the central and ring electrodes
24 [14]. However, it is at present unclear whether the effect of tDCS depends on the size and montage of
25 the electrodes. The most common explanation for anodal stimulation effects on learning is that it
26 facilitates depolarization of membrane potentials of cortical neurons, which are brought closer to their

intrinsic threshold for eliciting action potentials [15]. Such online effects (improvements seen during training) may eventually be important for influencing the long-term potentiation and depression underlying synaptic plasticity, possibly facilitating long-term memory formation [16]. As such, anodal tDCS of the M1 may boost motor learning outcomes, especially retention, in older adults.

In a randomized clinical trial, we found that when adding tDCS to writing training in people with Parkinson's disease, practice effects at 24-hour retention were enhanced compared to sham plus training [17]. However, in our recent meta-analysis involving older adults with and without Parkinson's disease, the added effects of tDCS seemed mixed and task-dependent [18]. Few studies examined the offline effect (improvements seen after training) of anodal M1 tDCS during MSL on retention in older adults [18]. They all applied CONV-tDCS montages and comprised small samples. Here, we examined the effects of two sessions of anodal tDCS of the M1 during MSL in a power-based sample of 52 older adults [BV1]. Additionally, we compared the effects of CONV- and HD-tDCS. Our primary hypothesis states that one session of tDCS+MSL will have a superior effect on 24-hour retention compared to sham+MSL. Our secondary hypothesis states that HD-tDCS will outperform CONV-tDCS when added to MSL for improving 24-hour retention.

Methods

This study was approved by the local Ethics Committee Research UZ/KU Leuven (S60893) and pre-registered in the Open Science Framework (OSF) (<https://osf.io/4yufh/>). Data collection occurred between July 2020 and March 2022.

Participants

Inclusion criteria were: (1) ≥ 50 years old; (2) right-handedness determined by the Edinburgh Handedness Inventory; (3) Mini Mental State Examination score ≥ 24 ; and (4) no other interfering disorders or implants.

Sample size was estimated at 52 subjects based on the literature (see [Supplementary Materials](#)). All subjects received a compensation of €50 for their participation and their study-related travel expenses were reimbursed.

Experimental design

[Figure 1A-B](#) shows our randomized double-blind sham-controlled study design. In accordance with [CONSORT requirements](#)^[JJOdx2], CONV- and HD-tDCS was applied to assess their added effect to motor sequence learning compared to sham. Participants underwent 4 sessions of either atDCS or sham. tDCS montages were manipulated within the atDCS and sham groups in a cross-over manner with a washout period of at least 7 days.

During the familiarization session (T0), the protocols were explained and clinical testing was conducted. After T0, an independent researcher carried out a concealed randomization procedure to assign subjects to the active atDCS or sham stimulation group. Randomization was performed in permuted blocks of four with gender and age (<65 or ≥65 years) as strata. The CONSORT flow diagram is shown in [Figure 2](#). During the first intervention session (T1), SRTT practice was combined with 20 minutes of atDCS or sham. Offline pre- and post-tests (tests without tDCS) of SRTT performance were also performed. At T2, 24 hours later, patients underwent offline retention tests of SRTT performance. Next, after a washout of 20 minutes, they received the same learning and stimulation protocol as at T1 to examine continued learning ability. The same protocol was repeated (T3 and T4) with a different tDCS montage in a randomized order ([Fig. 1A-B](#)).

Serial reaction time task

During each session, participants performed a SRTT (see [Figure 1C](#)). The fingers of the right hand, except the thumb, were assigned to four response keys (i.e. V, B, N and M). Four boxes were presented horizontally on a computer screen in front of the participant, with each box spatially corresponding to each of the four response keys. Participants were instructed to press the corresponding key as fast as

possible each time one of the boxes turned green. Key presses were grouped into blocks of 96 that were separated by 20-second breaks. Blocks were either randomly or sequentially ordered. Each sequenced block consisted of eight repetitions of the same 12-item sequence. Two sequences were practiced in counterbalanced order before and after cross-over: M-B-N-M-N-B-V-M-V-N-V-B (sequence 1) and B-M-V-B-V-M-N-B-N-V-N-M (sequence 2).

At T0, participants were familiarized with a random version of the SRTT. At T1/T4, participants completed two randomly and one sequentially ordered blocks as the offline pre-test. Next, participants engaged in online SRTT practice with atDCS or sham using the CONV- or HD-montage. They practiced 12 blocks of which all, except block 9, were sequentially ordered. Participants were made aware of the presence of a sequential pattern in the majority of blocks. After a 2-minute break (T2/T4), participants completed two sequenced blocks as the offline post-test. Twenty-four hours later, at T3/T6, the post-test was repeated to assess retention. After retention testing, a second session of online SRTT practice took place. Before the SRTT assessments at T1/T4 and T3/T6, participants performed a 10-minute PVT to assess vigilance (Fig. 1B).

The primary outcome was the 24-hour retention of SRTT performance gains in terms of mean reaction time. Successful retention was defined as the coexistence of: (1) a significant decrease in mean reaction time across sequenced blocks from pre (T1/T4) to post (T2/T5) and from pre (T1/T4) to retention (T3/T6) measurements; and (2) the absence of a significant increase in mean reaction time across sequenced blocks from post (T2/T5) to retention (T3/T6) measurements. Secondary outcome measures were SRTT performance during the initial learning and during the continued learning process. These were defined as the mean reaction times across sequenced blocks over the course of the first and second SRTT practice sessions. Another outcome is SRTT acquisition defined as the mean reaction times across sequenced blocks from pre (T1/T4) to post (T2/T5) tests. Analogously with aforementioned outcomes based on mean reaction times, the proportion of correct responses served as a measure of SRTT accuracy.

Transcranial direct current stimulation

During online practice, 20 minutes of atDCS or sham stimulation at 1mA was administered over the left M1, determined via TMS-procedures (see Supplementary Materials), using the CONV- or HD-montage (see Figure 1D). CONV-tDCS was applied via two electrodes of 5 cm x 7 cm (i.e. current density of 0.03 mA/cm²) with the cathode over the right supraorbital region. HD-tDCS consisted of a central anode with 2.5 cm diameter (i.e. current density of 0.2 mA/cm²) surrounded symmetrically by a ring-shaped cathode with 3.25 cm in between (see Figure 2). To ensure replication of electrode placement across sessions, participants wore an individualized head mold (see Supplementary Materials) before electrode placement. Contact impedance was monitored throughout stimulation (mean contact impedance was $6.3 \pm 1.4 \text{ k}\Omega$ [BV3] for the atDCS and $10.9 \pm 2.6 \text{ k}\Omega$ for the sham group). During active tDCS, the current was maintained constant with a 10-second ramp-up and -down. During sham, the current was ramped up and down over 30 seconds at the beginning and end of the stimulation period. Both participants and experimenter were blinded to stimulation type (atDCS or sham) by using the study mode of the tDCS stimulator (DC-Stimulator, NeuroConn GmbH, Germany). In this mode, a 5-digit number provided by an independent researcher encoded the stimulation type. Stimulation was administered by the same experimenter in all sessions. To monitor adverse effects, the tDCS questionnaire [19] was taken before and after stimulation. To check blinding, a visual analogue scale (VAS) assessing subjects' perceptions of how strongly they felt the stimulation was administered at the end of each session. After completing the last session, participants and experimenter were asked to guess whether they had received active tDCS or sham.

Data processing and statistical analysis

PVT reaction times (seconds) were extracted using Matlab R2020a (The MathWorks), while SRTT reaction times (milliseconds) and accuracy rates (%) using R software (v4.1.2; R Core Team 2022). The speed/accuracy trade-off was calculated by dividing reaction time (milliseconds) by accuracy (%). The higher this ratio, the more speed was prioritized over accuracy and vice versa.

Statistical analyses were conducted using SPSS software (version 28 SPSS, Inc., USA) with significance levels of $p < 0.05$. Parametric or non-parametric tests were performed after checking data distribution. Cross-tabulation was used to assess stimulation blinding of the participants and investigator based on stimulation perception (tDCS or sham). Mann-Whitney U tests were used to compare stimulation intensity as assessed by VAS scores between tDCS and sham groups. Group differences at baseline were checked with between-group statistics. Washout efficacy in terms of mean reaction times on the SRTT was insufficient based on a paired Wilcoxon signed ranks test ($p < 0.05$).

SRTT performance, side-effects and vigilance were analyzed with Linear Mixed Models with first-order autoregressive covariance structure of the repeated variable TIME. Significant interactions were analyzed by Bonferroni multiple comparison post-hoc tests. We controlled for within-subject differences and sequence effects by including participant and sequence as random effects. Hedges g based on pooled standard deviations was used to calculate effect sizes. Comparisons for SRTT performance included GROUP (tDCS or sham), TIME_[BV4]_[BV5] (pre, post, retention or post2 before and after crossover) and MONTAGE (CONV- or HD-tDCS) as fixed factors. For outcomes obtained during online practice and continued learning, fixed factors were GROUP, BLOCK (blocks 1-48) and MONTAGE. Side-effects and vigilance were analyzed for each session (session 1 or session 2 before and after crossover).

Results

Participants

Fifty-two people were randomized to either the atDCS ($n = 30$) or sham ($n = 22$) group (see Figure 2). Participant characteristics were comparable between groups (see Table 1).

There were no differences in objective vigilance between groups or experimental sessions (see Tables S1 and S2). However, participants of the sham group subjectively felt more fatigued than the atDCS group (see Tables S3 and S4). No differences in subjective fatigue were found between experimental sessions (see Tables S3 and S4).

Side-effects and blinding

We did not find evidence that the side-effects differed across group over time. The atDCS group reported significantly higher VAS scores of stimulation intensity [BV6] compared to the sham group after the intervention ($F(1, 52) = 7.553, p = 0.008, CI[0.272, 1.745]$). However, there were no significant differences in perceived stimulation for participants ($p = 0.397$), while significance was reached for the researcher ($p = 0.02$) (see Supplementary Table S3).

Offline effects[BV7]

SRTT performance at baseline did not differ between groups (GROUP*TIME: $F(7, 143) = 0.622, p = 0.737$). Retention intervals were 24 hours, except on 10 occasions (ranging from 60 minutes shorter to 30 minutes longer). The mean washout duration was 25.43 days (± 12.59 days). A significant main effect of TIME was found for the mean reaction time of the SRTT ($F(7, 143) = 66.257, p < 0.001$). Post[BV8][BV9]-hoc analysis showed that participants reacted significantly faster at the post-test (before crossover: $p < 0.001, CI[0.049; 0.117], g = 0.540$ and after crossover: $p < 0.001, CI[0.054; 0.122], g = 0.560$) and at the 24-hour retention test (before crossover: $p < 0.001, CI[0.036; 0.100], g = 0.443$ and after crossover: $p < 0.001, CI[0.025; 0.089], g = 0.363$) compared to baseline. No significant difference in mean reaction time was found from post- to retention-test (before crossover: $p < 0.001, CI[0.036; 0.100], g = -0.095$ and after crossover: $p < 0.001, CI[0.025; 0.089], g = -0.197$). After the continued practice session, a significant improvement in mean reaction time was found compared to performance at baseline, after the first practice session and before the continued practice session (all $p < 0.001$). No GROUP, GROUP*MONTAGE, GROUP*TIME, GROUP*TIME*MONTAGE effects were found (GROUP: $F(1, 93) = 0.270, p = 0.604$; GROUP*MONTAGE: $F(1, 93.028) = 0.057, p = 0.812$; GROUP*TIME: $F(7, 143) = 0.622, p = 0.737$; GROUP*TIME*MONTAGE: $F(7, 142.809) = 0.818, p = 0.574$). No effects were found on accuracy rates of the SRTT.

1 Online effects

2 During practice sessions, a significant main effect of BLOCK was found on mean reaction times
3 ($F(47,836) = 23.045, p < 0.001$). During the first practice session, post-hoc tests revealed that the
4 mean reaction time of sequenced blocks was significantly lower than of the random block (all $p < 0.05$).
5 Additionally, mean reaction time decreased significantly from block 1 to 2 (before crossover: $p = 0.037$,
6 $CI[0.000; 0.056]$, $g = 0.178$ and after crossover: $p = 0.041$, $CI[0.000; 0.055]$, $g = 0.178$). The lowest mean
7 reaction time of the first practice session equaled 517 milliseconds and was achieved during the second
8 block. [BV10] During the continued practice sessions, the mean reaction time of sequenced blocks was
9 significantly lower than of the random block (all $p < 0.001$). Additionally, the decrease in mean reaction
10 time from block 1 to 2 was significant before and nearly significant after crossover (before crossover:
11 $p < 0.001$, $CI[0.018; 0.073]$, $g = 0.286$ and after crossover: $p = 0.068$, $CI[0.000; 0.055]$, $g = 0.172$). The
12 lowest mean reaction time of the continued practice session was 465 milliseconds during the fourth
13 block. [BV11] Furthermore, participants reacted significantly faster during all blocks of the continued
14 practice session compared to the last block of the first practice session (all $p < 0.05$). Moreover, mean
15 reaction times throughout the continued practice session were lower than the second block of the first
16 practice session (i.e. block with lowest mean reaction time of first practice session), although not
17 always significantly. More specifically, significant differences were found with continued practice
18 blocks 2-7 and 12 before crossover, and with blocks 2-5 after crossover (all $p < 0.05$). We also found a
19 significant main effect of BLOCK on accuracy during (continued) practice sessions ($F(47,954) = 3.255$,
20 $p < 0.001$). Post-hoc tests showed significantly higher accuracy during the first 2 blocks of each
21 (continued) practice session versus the random block of the continued practice session (all $p < 0.05$).
22 Similarly, a significant main effect of BLOCK was found on the speed/accuracy trade-off ($F(47,847) =$
23 $19.559, p < 0.001$). The speed/accuracy trade-off of nearly all sequenced blocks was significantly
24 lower than of the random block of the (continued) practice sessions. In contrast to the mean reaction
25 times, the speed/accuracy trade-off did not lower significantly from block 1 to 2 (before crossover: p
26 $= 0.732$, $CI[-5.494; 60.896]$, $g = 0.169$ and after crossover: $p = 1.000$, $CI[-10.024; 56.366]$, $g = 0.141$),

1 due to accuracy changes in both directions. The lowest speed/accuracy trade-off was reached during
2 the second or third block of the (continued) practice sessions, i.e. 479 milliseconds.

3 Results of the per protocol analysis were performed with the same results (see Supplementary
4 materials – Table S1).

6 Discussion

7 We examined the added effect of CONV- and HD-tDCS of the M1 during SRTT practice on learning
8 outcomes in older adults. We assessed reaction times before, immediately after, 24 hours after and
9 during the two practice sessions of the SRTT supplemented by tDCS or sham. As the primary outcome,
10 we studied mean reaction times of the SRTT 24 hours after the first tDCS-enhanced practice session as
11 compared to sham. The linear mixed model revealed successful retention of the acquired SRTT
12 performance irrespective of receiving active or sham stimulation. Also, these effects were unaffected
13 by tDCS montage. These findings are contrary to our hypothesis that one session of tDCS+MSL would
14 have a positive added effect on 24-hour retention depending on tDCS montages. Analogously, the skill
15 acquisition process was unaffected by either tDCS or montage. Participants did react faster during all
16 blocks of the second practice session compared to the last block of the first practice session.

18 One stimulation session of the M1 may be insufficient to enhance retention in older adults, unlike in
19 people with Parkinson's disease as found in the recent study of Broeder et al. [17]. The engagement of
20 brain areas involved in MSL change with ongoing practice [2]. As M1 increasingly becomes important
21 in late MSL [2], it is possible that multiple sessions of tDCS are needed to effectively boost the
22 engagement of M1. This is also suggested by the lack of neurophysiological changes at the group level
23 after one session of tDCS preceded by MSL in older adults found by King et al. 2020 [20]. Two studies
24 examined the offline effect of multiple sessions of tDCS during MSL on retention in older adults [21,
25 22]. Greeley et al. 2022 found no tDCS benefit on the 24-hour retention of MSL reaction times after
26 two sessions of tDCS during MSL in small groups of 12 older adults each [22]. However, Dumel et al.

2018 reported the persistence of SRTT performance up to three months after five sessions of tDCS during MSL in a larger sample [21]. Noteworthy, their definition of retention reflected SRTT improvement from baseline to retention, independent of gains from baseline to post-test. In young adults, research demonstrated better retention after multiple sessions of M1 tDCS concurrently with MSL compared to sham [23-26]. Moreover, Ho et al. 2016 showed that repeated sessions of tDCS resulted in a cumulative increase in cortical excitability [27]. As such, the question remains whether multiple sessions of M1 tDCS lead to superior retention of MSL compared to sham in older adults.

Second, the stimulation intensity of 1 mA may have been too low to boost retention. A prior study showed enhanced LTP-like plasticity after applying 3mA tDCS to the M1 compared to lower intensities in older adults [28]. However, Ghasemian-Shirvan et al. 2023 did not find significant differences between 1, 2, and 3 mA tDCS of the M1 during MSL on neither acquisition nor consolidation outcomes in a large sample of elders [29]. In fact, they found no added effect of tDCS independent of stimulation intensity [29]. This is in line with our findings.

Another possibility is that our outcome measure may have lacked sensitivity and specificity to capture tDCS effects on retention. This is particularly important as the supplementary effects of tDCS on top of retention effects are possibly small and variable. To increase sensitivity and specificity, reaction times can be contrasted between sequentially and randomly ordered trials [30]. By unexpectedly replacing the sequence with random trials, the participant initially continues to reproduce the sequence [30]. This inflates the random reaction times, increasing the difference between sequential and random reaction times [30]. As such, this difference measure is more sensitive to sequence-specific learning [30]. Generally, sequence-specific improvements go together with a reduction in random reaction times, as participants also learn to make visuomotor associations [30]. As a result, the change in sequential reaction times alone provides a general measure of SRTT improvement less specific for sequence-specific learning [30]. Nevertheless, the disruptive effect of the random trial during SRTT

1 practice was indicative of sequence-specific learning in our sample. By contrasting sequential and
2 random reaction times, influence of possible [contaminators](#)^[BV12] can also be minimized. Nevertheless,
3 there is no reason to assume that this was the case, as objective vigilance did not differ between
4 groups. The higher subjective feeling of fatigue in the sham group does not seem to have influenced
5 our results, as no group differences in SRTT reaction times were found.

6
7 The fourth reason for the absence of a significant difference between tDCS and sham on retention is
8 the possibility of placebo effects induced by sham. Sham is assumed to provide the same sensations
9 as tDCS (e.g. itching), thereby aiming to optimize participant blinding [31]. In our study, participants
10 were unaware of the allocated intervention, indicating successful participant blinding, despite higher
11 VAS scores of stimulation intensity after active tDCS compared to sham. In line with our findings,
12 Ambrus et al. 2012 demonstrated sham to be reliable at 1 mA for tDCS naïve subjects, but not for
13 experienced subjects [32]. Experienced subjects were more likely to correctly differentiate stimulation
14 from sham [32]. In our study, only four participants were experienced with tDCS of which all guessed
15 their group allocation correctly (tDCS: n=3 and sham: n=1). Nevertheless, participant blinding was
16 achieved at the group level.

17
18 Another explanation is increased variability in the tDCS response by applying HD-tDCS and CONV-tDCS
19 montages. A full understanding of the tDCS mechanisms underlying behavioral effects is lacking.
20 However, it is assumed that several mechanisms contribute. As the current passes through the scalp,
21 skull and cerebrospinal fluid, its electric field is attenuated when reaching the cortex. This field can
22 polarize the membrane potentials of cortical neurons. This is considered the transcranial mechanism.
23 Van Boekholdt et al. 2020 also proposed a peripheral mechanism [15]. As tDCS electrodes are placed
24 on the skin, peripheral nerves are exposed to electric fields that are sufficiently strong to initiate action
25 potentials in cranial and cervical nerves [13]. Subsequent locus coeruleus activation leads to the
26 release of norepinephrine, causing behavioral effects by increasing cortical excitability and synaptic

1 plasticity [15]. We assume that HD-tDCS produces a more focal though superficial current than CONV-
2 tDCS, because of its smaller inter-electrode distance [33]. Therefore, we suggest that the relative
3 contribution of the peripheral mechanism may be larger for HD- than CONV-tDCS. As such, pooling HD-
4 and CONV-montages may have led to increased variability in the behavioral tDCS response in this
5 study.

6
7 Finally, it is possible that the effectiveness of tDCS is impacted by the changes in brain architecture and
8 cortical electrophysiology in older age. Evidence suggests that the thickness of the skull and
9 cerebrospinal fluid, which impact the magnitude of the electric field at the cortical level [34, 35],
10 change with age [36, 37]. These changes could have attenuated electric fields in older adults. The
11 electrophysiological changes in cortical neurons in older age, may also influence the effectiveness of
12 tDCS [38]. Stagg et al. 2011 showed that the responsiveness of the neurotransmitter gamma-
13 aminobutyric acid, modified by M1-tDCS, correlated with motor learning capacity in healthy [39]. As a
14 result of these changes, tDCS may elicit a differential neural response and therefore be less effective
15 at the group level.

16 17 **Strengths and limitations**

18 As a response to concerns that were raised regarding the methodological robustness of previous tDCS
19 research [8], this study was a preregistered randomized sham-controlled trial based on an a priori
20 sample size calculation. Our pre-specified hypotheses and outcomes captured the 'offline' effects of
21 tDCS added to practice in each learning phase. Moreover, objective vigilance did not influence learning
22 effects. TDCS tolerability and perceived assignment to either the tDCS or sham group showed
23 successful participant but not researcher blinding. We reported this research using the checklist
24 proposed by Buch et al. 2017 as a guideline.

25 26 **Future research**

Future research may focus on the offline effect of multiple sessions of tDCS during MSL on long-term retention and other consolidation outcomes. For interpretation purposes, quantitative assessments of vigilance, tDCS tolerability and perceived assignment to either the tDCS or sham group should always be performed [40]. A measure to reduce stimulation sensations and thus the role of expectations is the application of a topical anesthetic to block peripheral nerve input [41]. An additional control group receiving training without tDCS or sham could be included in order to gain insight on possible placebo effects.

Conclusions

We conclude that there is no surplus value of adding tDCS during motor sequence learning in older adults. We did find that older people benefited from a second practice session to improve their motor learning. Future research should focus on the effects of multiple sessions of tDCS during motor sequence learning on long-term retention and other consolidation outcomes. As learning impairments are more pronounced in people with Parkinson's disease compared to healthy, benefit from stimulation should be studied in this population.

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Author contributions

1. Research project: A. Conception, B. Organization, C. Execution; 2. Data: A. Acquisition, B. Processing; 3. Statistical Analysis: A. Design, B. Execution, C. Review and Critique; 4. Manuscript: A. Writing of the first draft, B. Review and Critique

BV: 1A, 1B, 1C, 2A, 2B, 3A, 3B, 4A

SB: 1A, 1B, 1C, 2A, 2B, 3A, 3B, 3C, 4B

MG: 1A, 3A, 4B

1 EN: 1A, 1B, 1C, 3A, 3C, 4B

2 JO: 1A, 1B, 1C, 3A, 3C, 4B

3 AN: 1A, 1B, 1C, 3A, 3C, 4B

4

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8

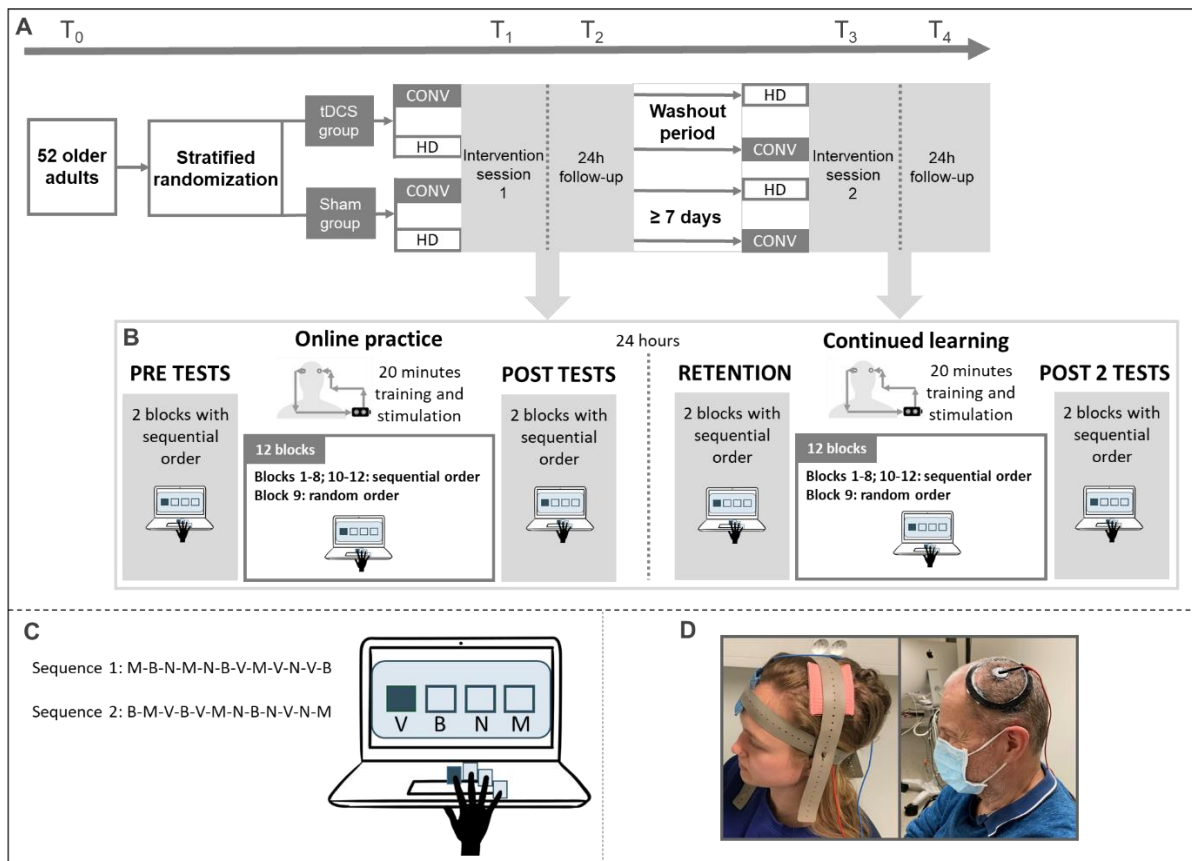


Figure 1. Study design. A) Timeline; B) Intervention and assessments; C) Serial reaction time task; and C) Conventional (left) and high-definition (right) transcranial direct current stimulation montages. CONV: conventional; HD: high-definition; tDCS: transcranial direct current stimulation.

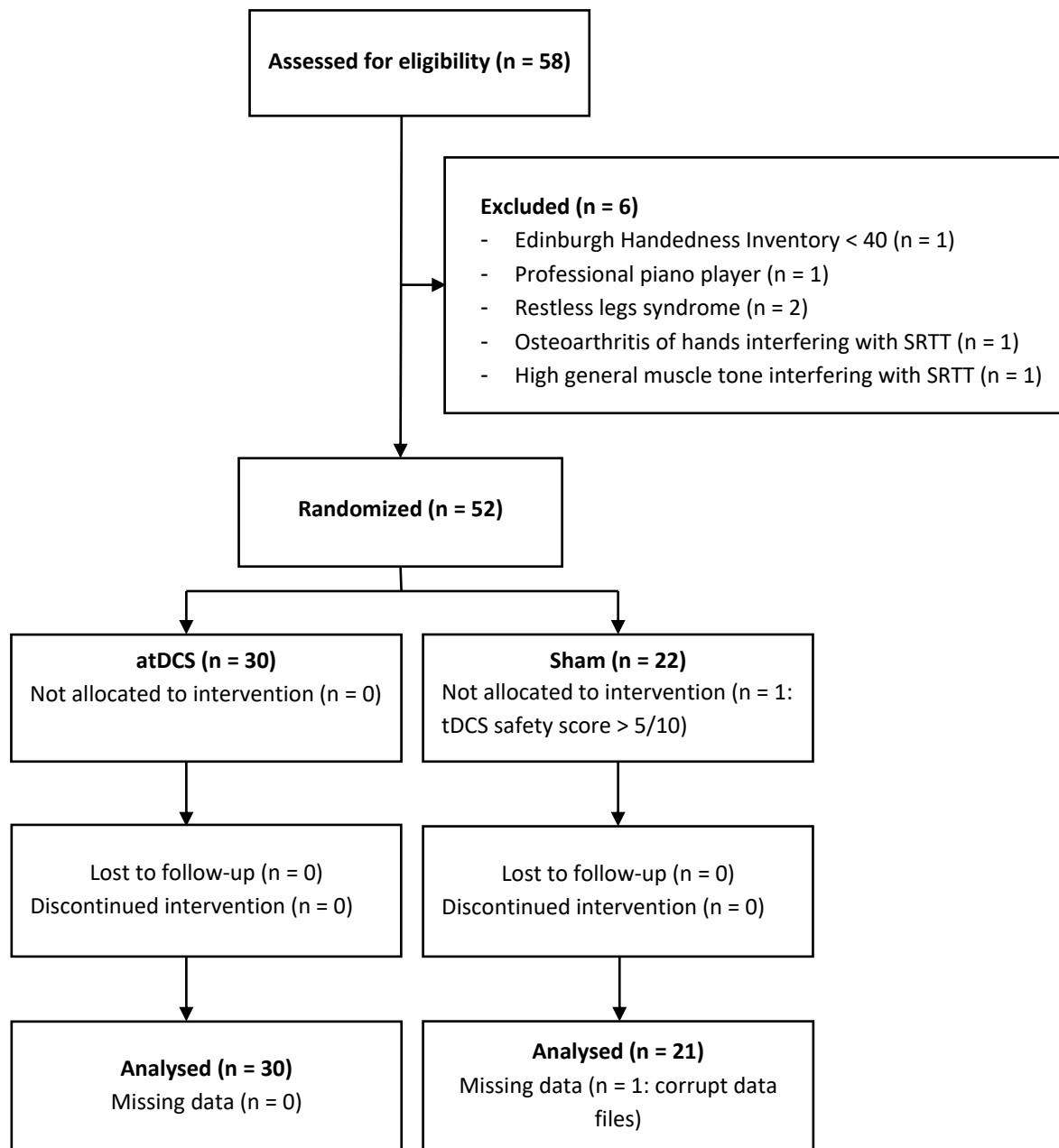


Figure 2. Consort flow diagram.

Abbreviations: SRTT, serial reaction time task; tDCS, transcranial direct current stimulation.

1

Table^[BV13] **1.** Participant characteristics ^a

	atDCS (n = 30)	Sham (n = 22)	p-value ^b
Age (years)	69.03 ± 7.49	67.55 ± 8.82	0.52
Gender (female, male)	(15,15)	(10,12)	0.79
MoCA (0-30)	27.03 ± 2.03	26.73 ± 2.14	0.60
TMT-A (seconds)	27.50 (20.00, 52.00)	29.00 (16.00, 57.00)	0.36
TMT-A (errors)	0.00 (0.00, 2.00)	0.00 (0.00, 3.00)	0.39
TMT-B (seconds)	68.00 (42.00, 117.00)	79.00 (35.00, 110.00)	0.63
TMT-B (errors)	0.00 (0.00, 2.00)	0.00 (0.00, 4.00)	0.22
PSQI (0-21)	4.40 ± 2.34	4.82 ± 2.63	0.55
HADS Anxiety (0-21)	3.50 (0.00, 15.00)	3.00 (0.00, 12.00)	0.93
HADS Depression (0-21)	1.00 (0.00, 8.00)	2.00 (0.00, 10.00)	0.40
DEXTQ-24 (24-96)	24.00 (24.00, 26.00)	24.00 (24.00, 25.00)	0.37

Abbreviations: HADS, Hospital Anxiety and Depression Scale; MoCA, Montreal Cognitive Assessment.

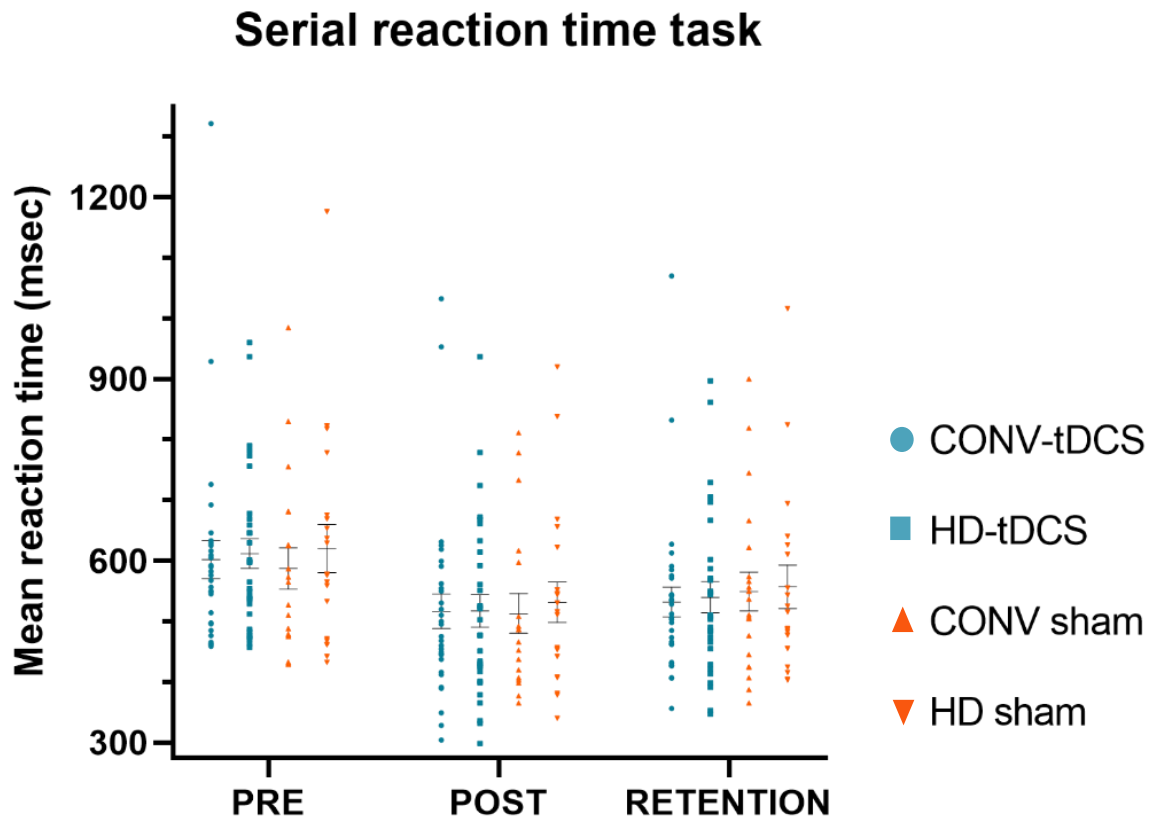
^a Results are presented as the mean (± standard deviation) for normally distributed variables and as the median (minimum, maximum) for non-normally distributed variables.

^b Groups significantly different at P <.05.

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3 **Figure 3.** Mean reaction time (msec) on the serial reaction time task before, immediately after and 24
4 hours after conventional and high-definition tDCS and sham stimulation. Data are presented as
5 individual data points with group means ($\pm$ standard error of the mean). CONV-tDCS: conventional
6 tDCS; HD-tDCS: high-definition tDCS; tDCS: transcranial direct current stimulation.

7

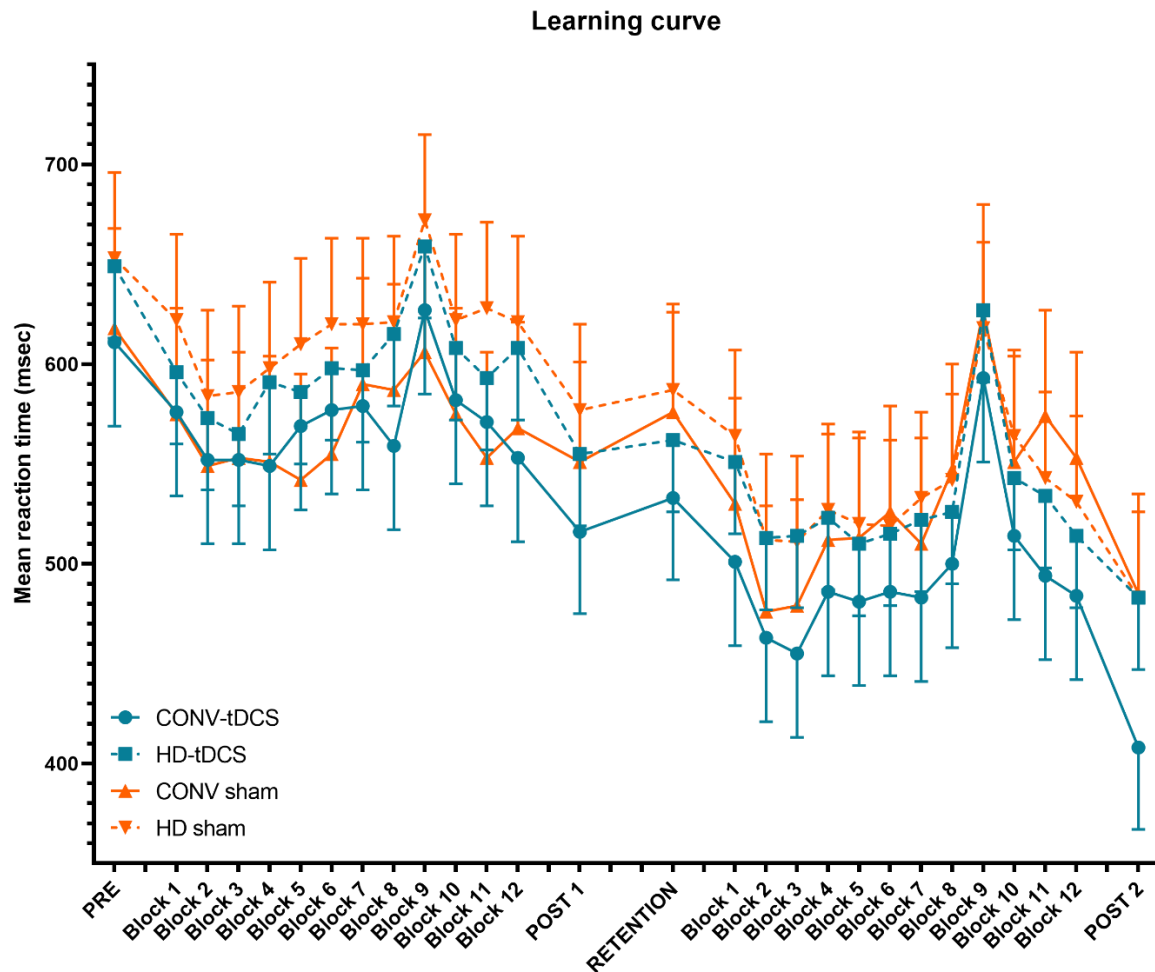


Figure 4. Mean reaction times (msec) on the serial reaction time task for conventional and high-definition tDCS and sham stimulation. Data are presented as group means ($\pm$ standard error of the mean).

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Supplementary materials

Sample size calculation

Performing an a-priori sample size calculation for this study is difficult, as no previous comparison exists on the behavioral effects of HD-tDCS and CONV-tDCS using the same SRTT and primary outcome measure. Our best estimate of the target sample size is based on the behavioral effects of HD-tDCS compared to CONV-tDCS in healthy adults (calculated effect size f of 0.12) [42]. This was derived from the effect of active anodal tDCS of M1 compared to sham-tDCS combined with a finger sequence learning task in healthy older adults [43]. Using G*Power software, a target sample of 24 participants per group was determined to reach a power of 80% with an alpha of 5% in a general linear model design. In anticipation of an unforeseen drop-out rate of 10%, a total of 26 participants per group will be included. Recruitment will stop when we reach our target sample size.

Transcranial magnetic stimulation

TMS-procedures

Single and paired-pulse TMS (Magstim BiStim2, UK, figure-of-eight coil, 70 mm diameter) were delivered at the optimal scalp position (hotspot) to elicit motor responses in the right first dorsal interosseous (FDI) muscle (surface electromyography: Bagnoli-4, Delsys Inc., USA). During the baseline session, the hotspot for the FDI of the left hemisphere was determined.[44] To ensure a consistent delivery of TMS stimuli to the same spot at the primary motor cortex (M1) across sessions, a customized head mold of aqua-adaptable splinting material (Aquaplast-T Solid, Sammons Preston Polyon, Cedarburg) with a marked orthogonal coordinate system was created for each participant.[45]

Data processing cortical excitability data

After determining the background EMG (i.e. root mean square EMG signal during the 50 ms period immediately preceding the onset of the TMS pulse) and MEP peak-to-peak values, all individual MEP

trials were screened and excluded if they met one of the following criteria: (i) background EMG > 5 μ V 5; (ii) background EMG > mean +2.5 standard deviations (SD) for trials in rest (iii) background EMG > mean +2.5 SDs and/or < mean -2.5 SDs for active trials or (iv) outlier MEPs with a peak-to-peak amplitude > $Q3+1.5*(Q3-Q1)$ or < $Q1-1.5*(Q3-Q1)$ (with Q1 = first quartile and Q3 = third quartile).[46]

Following cleaning, 94.1 ± 1.3 % of the trials were included for further statistical analysis. All TMS outcomes were calculated as the average MEPs measured pre and post intervention. For MEP amplitude, the post/pre ratios were calculated to assess intervention effects (i.e. values above or below 1 indicated an increase or decrease in excitability). SICI was calculated as a percentage of the unconditioned MEP peak-to-peak amplitudes with values closer to 0% representing more intracortical inhibition.

Washout efficacy

Table S1: Mean reaction time at baseline of the serial reaction time task before and after crossover per group

SRTT			
Mean reaction time (milliseconds)	T1 - pre	T3 - pre	p-value
atDCS	635 $\pm$ 176	579 $\pm$ 120	1.000
sham	638 $\pm$ 183	587 $\pm$ 145	1.000

Values represent means $\pm$ standard deviations. SRTT = Serial Reaction Time Task.

Objective and subjective vigilance

Table S2: [BV14]Objective vigilance per group for each experimental session

PVT (milliseconds)	T1	T2	T3	T4
atDCS	319 $\pm$ xxx	314 $\pm$ xxx	308 $\pm$ xxx	306 $\pm$ xxx
sham	328 $\pm$ xxx	323 $\pm$ xxx	319 $\pm$ xxx	320 $\pm$ xxx

Values represent means $\pm$ standard deviations. Vigilance was assessed at the beginning of each experimental session with the Psychomotor Vigilance Test (referentie). Participants completed 4 experimental sessions. PVT = Psychomotor Vigilance Test.

Table S3: Linear mixed model analysis of objective vigilance with group (atDCS vs. sham) and session (1-4) as fixed and subject number as random factors

	Main effect of group	Main effect of session	Group by session interaction
PVT (milliseconds)	$F_{(1)} = 1.28, p = .264$	$F_{(3)} = 1.125, p = .341$	$F_{(3)} = 0.071, p = .976$

PVT = Psychomotor Vigilance Test.

Table S4: Fatigue per group for each experimental session

VAS-scores	T1	T2	T3	T4
atDCS	8.72 ± 1.18	8.61 ± 1.65	8.53 ± 1.47	9.05 ± 0.76
sham	8.16 ± 1.01	8.59 ± 0.90	7.64 ± 2.08	7.89 ± 2.47

Values represent means $\pm$ standard deviations. Fatigue was assessed at the beginning of each experimental session with Visual Analogue Scales (0: severe fatigue; 10: no fatigue). Participants completed 4 experimental sessions. VAS = Visual Analogue Scale.

Table S5: Linear mixed model analysis of fatigue with group (atDCS vs. sham) and session (1-4) as fixed and subject number as random factors

	Main effect of group	Main effect of session	Group by session interaction
VAS-scores	$F_{(1)} = 5.76, p = .020$	$F_{(3)} = 2.21, p = .090$	$F_{(3)} = 1.87, p = .137$

VAS = Visual Analogue Scale.

Blinding

Table S6. Results on the assessment of stimulation blinding

Stimulation perception	Answer = atDCS	Answer = Sham	Answer = ?
Participants	tDCS group	21	6
	Sham group	12	8
Researcher	tDCS group	21	9
	Sham group	5	16

1 *Motor sequence learning*

2 **Table S7.** Offline assessment of the serial reaction time task in terms of mean reaction time (standard
3 deviation)

Assessment time points	tDCS montage	atDCS group	sham group	atDCS vs. sham p-value
PRE	CONV	616 (231)	618 (160)	0.912
	HD	649 (130)	653 (204)	0.954
	p-value	0.487	0.602	
POST	CONV	516 (169)	520 (112)	0.591
	HD	555 (163)	577 (165)	0.690
	p-value	0.482	0.695	
RETENTION	CONV	533 (169)	544 (110)	0.515
	HD	569 (137)	587 (182)	0.665
	p-value	0.596	0.867	
POST 2	CONV	408 (130)	453 (152)	0.244
	HD	483 (175)	485 (152)	0.998
	p-value	0.175	0.981	

4 Abbreviations: ACC = accuracy rate; atDCS = anodal transcranial direct current stimulation; RT= mean reaction
5 time. Alpha = 0.05

6

1 **Table S8.** Online assessment of the serial reaction time task in terms of mean reaction time (standard
2 deviation)

Intervention session	SRTT block	tDCS montage	atDCS group	sham group	atDCS vs. sham p-value
Session 1	Block 1 ^S	CONV	576 (155)	575 (139)	0.988
		HD	596 (139)	622 (197)	0.643
		p-value	0.723	0.495	
	Block 2 ^S	CONV	552 (148)	549 (165)	0.962
		HD	573 (132)	584 (172)	0.847
		p-value	0.705	0.609	
	Block 9 ^R	CONV	627 (213)	606 (123)	0.764
		HD	659 (147)	672 (175)	0.820
		p-value	0.563	0.343	
	Block 10 ^S	CONV	583 (174)	575 (143)	0.917
		HD	608 (168)	621 (185)	0.813
		p-value	0.641	0.500	
	Block 12 ^S	CONV	553 (166)	568 (150)	0.833
		HD	608 (193)	621 (198)	0.824
		p-value	0.327	0.443	
Session 2	Block 1 ^S	CONV	501 (134)	530 (143)	0.670
		HD	551 (151)	564 (170)	0.808
		p-value	0.371	0.615	
	Block 2 ^S	CONV	463 (115)	476 (109)	0.850
		HD	513 (147)	512 (153)	0.983
		p-value	0.367	0.599	
	Block 9 ^R	CONV	593 (158)	627 (200)	0.616
		HD	627 (146)	618 (124)	0.872
		p-value	0.537	0.899	
	Block 10 ^S	CONV	515 (165)	551 (199)	0.587
		HD	543 (186)	564 (137)	0.717
		p-value	0.605	0.855	
	Block 12 ^S	CONV	484 (158)	553 (241)	0.311
		HD	514 (180)	531 (126)	0.766
		p-value	0.591	0.749	

3 Abbreviations: ACC = accuracy rate; atDCS = anodal transcranial direct current stimulation; R = randomly ordered;
4 RT = mean reaction time; S = sequentially ordered.
5 Alpha = 0.05

1 *Per protocol analysis*

2 Protocol deviations were impedance $>20k\Omega$ on 4 occasions and interruption of the 20 minutes
3 stimulation period on 5 occasions. Participants were not informed about these deviations. Seven
4 participants of the sham group and two participants of the active tDCS group were affected. These
5 deviations were not expected to influence the outcomes of the sham group. Consequently, data from
6 two participants in the active tDCS group were excluded from the per protocol analysis, either before
7 ($n=1$) or after ($n=1$) crossover.

8 **Offline effects**

10 No significant differences in SRTT performance at baseline were present between groups
11 (GROUP*TIME: $F(7, 133) = 1.408, p = 0.207$). There was no significant difference in baseline SRTT
12 performance before and after crossover within groups (active tDCS group: $p = 1.000$, $CI[-0.046; 0.150]$,
13 $g = xxx$ and sham group: $p = 1.000$, $CI[-0.046; 0.150]$, $g = xxx$).

14 A significant main effect of TIME on the mean reaction time of the SRTT was found ($F(7, 133) =$
15 $61.848, p < 0.001$). Post-hoc analysis showed that participants reacted significantly faster at the post-
16 test compared to baseline (before crossover: $p < 0.001$, $CI[0.039; 0.110]$, $g = xxx$ and after crossover: p
17 < 0.001 , $CI[0.056; 0.126]$, $g = xxx$). Additionally, participants reacted significantly faster at the 24-hour
18 retention test compared to baseline (before crossover: $p < 0.001$, $CI[0.033; 0.100]$, $g = xxx$ and after
19 crossover: $p < 0.001$, $CI[0.020; 0.089]$, $g = xxx$). No significant difference in mean reaction time was
20 found from post- to retention-test (before crossover: $p = 1.000$, $CI[-0.043; 0.027]$, $g = xxx$ and after
21 crossover: $p = 0.040$, $CI[-0.073; -0.001]$, $g = xxx$)^[BV15]. After the continued practice session, a significant
22 improvement in mean reaction time was found compared to performance at baseline , after the first
23 practice session and before the continued practice session (all $p < 0.001$). Neither GROUP nor
24 GROUP*TIME effects were found (GROUP: $F(1, 93) = 0.270$, $p = 0.604$; GROUP*TIME: $F(7, 143) = 0.622$,

p = 0.737). These results showed 24-hour retention of acquired SRTT performance and successful continued learning irrespective of tDCS. No effects were found on accuracy rates of the SRTT. When analyzing SRTT performance per tDCS montage, neither GROUP nor GROUP*TIME effects were found for mean reaction times (CONV-tDCS - GROUP: $F(1, 46.743) = 0.286$, $p = 0.595$; GROUP*TIME: $F(7, 68.936) = 1.159$, $p = 0.338$; HD-tDCS - GROUP: $F(1, 46.056) = 0.042$, $p = 0.838$; GROUP*TIME: $F(7, 55.777) = 0.198$, $p = 0.985$). When taking into account age, the results showed a nearly significant GROUP*TIME*AGE interaction for mean reaction times of the SRTT ($F(22, 42.409) = 1.651$, $p = 0.080$). When analyzing the subset of participants older than 65 ($n = 39$), a significant main effect of TIME on the mean reaction time of the SRTT was found ($F(7, 103.467) = 50.060$, $p < 0.001$). Neither GROUP nor GROUP*TIME effects were found (GROUP: $F(1, 66.898) = 0.347$, $p = 0.558$; GROUP*TIME: $F(7, 103.467) = 0.728$, $p = 0.649$).

Online effects

During practice sessions, a significant main effect of BLOCK was found on mean reaction times ($F(47, 836) = 23.045$, $p < 0.001$). During the first practice session, post-hoc tests revealed that the mean reaction time of sequenced blocks was significantly lower than of the random block (all $p < 0.05$). Additionally, mean reaction time decreased significantly from block 1 to 2 (before crossover: $p = 0.037$, $CI[0.000; 0.056]$, $g = xxx$ and after crossover: $p = 0.041$, $CI[0.000; 0.055]$, $g = xxx$). The lowest mean reaction time of the first practice session equaled 517 milliseconds and was achieved during the second block. During the continued practice sessions, the mean reaction time of sequenced blocks was significantly lower than of the random block (all $p < 0.001$). Additionally, the decrease in mean reaction time from block 1 to 2 was significant before and nearly significant after crossover (before crossover: $p < 0.001$, $CI[0.018; 0.073]$, $g = xxx$ and after crossover: $p = 0.068$, $CI[0.000; 0.055]$, $g = xxx$). The lowest mean reaction time of the continued practice session was 465 milliseconds during the fourth block. Furthermore, participants reacted significantly faster during all blocks of the continued practice session compared to the last block of the first practice session (all $p < 0.05$). Moreover, mean reaction

times throughout the continued practice session were lower than the second block of the first practice session (i.e. block with lowest mean reaction time of first practice session), although not always significant. More specifically, significant differences were found with continued practice blocks 2-7 and 12 before crossover, and with blocks 2-5 after crossover (all $p < 0.05$). We also found a significant main effect of BLOCK on accuracy during (continued) practice sessions ($F(47,954) = 3.255, p < 0.001$). Post-hoc tests showed significantly higher accuracy during the first 2 blocks of each (continued) practice session versus the random block of the continued practice session (all $p < 0.05$). These results point towards sequence-specific learning strengthened by a second practice session independent of adding tDCS. Similarly, a significant main effect of BLOCK was found on the speed/accuracy index ($F(47,847) = 19.559, p < 0.001$). The speed/accuracy index of nearly all sequenced blocks was significantly lower than of the random block of the (continued) practice sessions. In contrast to the mean reaction times, the speed/accuracy index did not lower significantly from block 1 to 2 (before crossover: $p = 0.732$, $CI[-5.494; 60.896]$, $g = xxx$ and after crossover: $p = 1.000$, $CI[-10.024; 56.366]$, $g = xxx$), due to accuracy changes in both directions. The lowest speed/accuracy index was reached during the second or third block of the (continued) practice sessions, i.e. 479 milliseconds. Analyses of transformed outcomes were performed with the same results (see Supplementary materials – Table S1). Results of the per protocol analysis were performed with the same results (see Supplementary materials – Table S1).